Tokenized Option Ladder: Mathematical Formulation

1 Setup and the Splitting Primitive

Maturity and clearing. A position is opened against collateral and settled at a single maturity m. At maturity an oracle posts the terminal underlying price X_m *once*; every tranche then clears to a closed-form cash (or collateral) amount. The oracle is therefore needed only at this clearing instant, not during the life of the position.

Two collateral states. Every construction starts from one of two collateral states: one unit of the underlying T (terminal value X_m), or cash C (in the put-side base split $C = S_0$).

The splitting primitive. At a chosen strike S, any collateral position splits into an *obligation residual* P and a *right* N,

$$\text{Collateral} \;=\; \underbrace{P}_{\text{residual}} + \underbrace{N}_{\text{right}}, \qquad P(X_m) + N(X_m) = (\text{terminal value of the collateral}) \;\; \forall X_m.$$

This conservation identity is what makes every tranche *fully collateralized*: the pieces always sum back to the collateral that backs them.

Tranche-address notation (the P/N tree). The suffix on each label is the path taken through the recursion. Starting from the collateral, each split sends the residual to a leaf (P) and keeps splitting the right (N):

$$\begin{aligned} T \;&\to \{\, \text{CALL}_P \text{ (leaf)},\; \text{CALL}_N \,\} \\ &\to \text{CALL}_N \text{ splits into } \{\, \text{CALL}_{NP} \text{ (leaf)},\; \text{CALL}_{NN} \,\} \\ &\to \text{CALL}_{NN} \text{ splits again } \;\to\; \cdots \end{aligned}$$

The strike ladder *is* this recursion; the running index i simply flattens the repeated-N path, so $\text{CALL}_{NP,\,i}$ is the residual produced at the i-th split.

Numeraire. Payoffs are written in cash / numeraire value by default; collateral-unit forms (underlying-T units, cash units) are given where useful.

Scope. This section is a payoff *decomposition*, not a pricing model. Premiums are set by the market; the math here only guarantees that the tranches partition the collateral's terminal value exactly.

Notation

Symbol	Meaning
X	Underlying asset price.
X_m	Terminal underlying price at maturity m, expressed in cash/numeraire units.

T	One unit of underlying collateral. Its terminal cash value is X_m.
C	Cash collateral. In the put-side base split, $C = S_0$.
S	Strike price.
S_0	Strike price at the first split (the base strike).
S_i	The i-th strike price, where $i = 1, \dots, n$ indexes the rungs of the ladder. Call ladder: $S_0 < S_1 < \cdots < S_n$; put ladder: $S_0 > S_1 > \cdots > S_n$.
P	Obligation residual.
N	Right component: call right or put right.
CALL_i	Tokenized call position struck at S_i (a token, not a traded option); terminal value $\max(0, X_m - S_i)$.
PUT_i	Tokenized put position struck at S_i (a token, not a traded option); terminal value $\max(0, S_i - X_m)$.
$+\mathrm{CALL}_i$	Long the call token CALL_i.
$-\mathrm{CALL}_i$	Short the call token CALL_i.
$+\mathrm{PUT}_i$	Long the put token PUT_i.
$-\mathrm{PUT}_i$	Short the put token PUT_i.
CC	Covered call (strategy): hold the underlying and write a call against it, capping the upside above the strike in return for the premium.
CCR_i ($\mathrm{CALL\_P}_i$)	Covered-call residual token at rung i: the residual leg of the covered call CC applied at strike S_i.
CSP	Cash-secured put (strategy): hold cash and write a put fully backed by that cash, taking on the obligation to buy the underlying below the strike.
CSPR_i ($\mathrm{PUT\_P}_i$)	Cash-secured-put residual token at rung i: the residual leg of the cash-secured put CSP applied at strike S_i.
BCS	Bull call spread (strategy): buy a lower-strike call and sell a higher-strike call, for a bounded bullish payoff between the two strikes.
BCSR_i ($\mathrm{CALL\_NP}_i$)	Bull-call-spread residual token at rung i: the residual leg of the bull call spread BCS between strikes S_{i-1} and S_i.
BPS	Bear put spread (strategy): buy a higher-strike put and sell a lower-strike put, for a bounded bearish payoff between the two strikes.
BPSR_i ($\mathrm{PUT\_NP}_i$)	Bear-put-spread residual token at rung i: the residual leg of the bear put spread BPS between strikes S_{i-1} and S_i.

All formulas below are terminal payoff formulas unless explicitly written in collateral-token units.

2 Route Strategy Map

The route strategy map is the index for everything that follows. From each collateral state it shows the base split into a residual and a right, and the recursive split of the right along the strike ladder into the vertical-spread residuals — enumerating the full menu of tranches. Each strategy in Sections 3–4 is a labeled route on this map; the wheel of Section 5 is the cycle that connects the two collateral states.

*[**TODO:** Route strategy map figure: collateral state → side (call / put) → base split → recursive split*

→ tranche menu ($\mathrm{CALL}_P, \mathrm{CALL}_N, \mathrm{CALL}_{NP}, \mathrm{CALL}_{NN}$; $\mathrm{PUT}_P, \mathrm{PUT}_N, \mathrm{PUT}_{NP}, \mathrm{PUT}_{NN}$)*. Overview / index figure; the per-strategy diagrams in Sections 3–4 are zoom-ins of it.]*

3 Call Side

Begin with one unit of underlying collateral T, whose terminal value is X_m. At the base strike S_0 the underlying splits into an obligation residual and a right:

$$T = (T - \mathrm{CALL}_{S_0}) + \mathrm{CALL}_{S_0}, \tag{1}$$

$$T = \mathrm{CALL}_{P,\,S_0} + \mathrm{CALL}_{N,\,S_0}. \tag{2}$$

The residual $\mathrm{CALL}_{P,\,S_0}$ is the covered-call position (Section 3.2); the right $\mathrm{CALL}_{N,\,S_0}$ is the long call (Section 3.1), which is split further along the ladder into bull call spreads (Section 3.3). Every call-side tranche conserves to X_m.

3.1 Long Call (CALL_N)

How it works. The long call is the *right* half of the base split: a tokenized call position struck at S_0. Its holder owns the uncapped upside above S_0 and nothing below, while the matching residual (CALL_P) keeps the underlying that collateralizes it. It is the bullish, long-exposure tranche, and the only leg that splits further down the ladder.

Splitting. From the base split $T = \mathrm{CALL}_{P,\,S_0} + \mathrm{CALL}_{N,\,S_0}$, the long call is the right component

$$\mathrm{CALL}_{N,\,S_0} = \mathrm{CALL}_{S_0}. \tag{3}$$

Clearing (settlement).

$$\mathrm{CALL}_{N,\,S_0}(X_m) = (X_m - S_0)^+ = \max(0, X_m - S_0). \tag{4}$$

In underlying-collateral units, when $X_m > 0$:

$$\mathrm{CALL}^T_{N,\,S_0}(X_m) = \max\left(0, 1 - \frac{S_0}{X_m}\right) T. \tag{5}$$

*[**TODO:** Split + clearing diagram for the long call (zoom-in of the wheel: the* CALL_N *right leg).]*

*[**TODO:** Settlement chart:* $\mathrm{CALL}_{N,\,S_0}(X_m)$ *vs. underlying USD price* X_m *(flat* 0 *up to* S_0*, then* $45°$*).]*

*[**TODO:** Settlement table for this tranche across* X_m *(the* CALL_N *column of the combined table in Section 3.4).]*

3.2 Covered Call (CALL_P / CCR)

How it works. The covered call is the *residual* half of the base split: the underlying minus the call that was sold off. Its holder keeps the asset below S_0 and is converted to cash S_0 above it — the classic covered-call writer's position, with upside capped at the strike.

Splitting. From the base split,

$$\mathrm{CALL}_{P,\,S_0} = T - \mathrm{CALL}_{S_0}. \tag{6}$$

Clearing (settlement).

$$\mathrm{CALL}_{P,\,S_0}(X_m) = \mathrm{CCR}_{S_0}(X_m) = X_m - (X_m - S_0)^+ = \min(X_m, S_0). \tag{7}$$

Settlement semantics: if $X_m \geq S_0$ the call is assigned and the residual converts the underlying into cash S_0; if $X_m < S_0$ it retains the underlying T, worth X_m. In underlying-collateral units, when $X_m > 0$:

$$\mathrm{CALL}^T_{P,\,S_0}(X_m) = \min\left(1, \frac{S_0}{X_m}\right) T. \tag{8}$$

Conservation with the long call:

$$\mathrm{CALL}_{P,\,S_0}(X_m) + \mathrm{CALL}_{N,\,S_0}(X_m) = \min(X_m, S_0) + (X_m - S_0)^+ = X_m. \tag{9}$$

*[**TODO:** Split + clearing diagram for the covered call (the CCR node: $T \rightarrow$Cash when $X_m \geq S$, keep T when $X_m < S$).]*

*[**TODO:** Settlement chart: $\mathrm{CALL}_{P,\,S_0}(X_m)$ vs. underlying USD price X_m ($45°$ then capped flat at S_0).]*

*[**TODO:** Settlement table for this tranche across X_m (the CALL_P column of the combined table in Section 3.4).]*

3.3 Bull Call Spread (CALL_{NP} / BCSR)

How it works. Recursively split the call right at the next rung. For a call ladder with increasing strikes

$$S_0 < S_1 < \cdots < S_n,$$

each lower-strike call splits into a capped bull call spread residual plus the higher-strike call that is carried further down the ladder.

Splitting. For $i = 1, \ldots, n$:

$$\mathrm{CALL}_{N,\,S_{i-1}} = (\mathrm{CALL}_{S_{i-1}} - \mathrm{CALL}_{S_i}) + \mathrm{CALL}_{S_i}, \tag{10}$$

$$\mathrm{CALL}_{N,\,S_{i-1}} = \mathrm{CALL}_{NP,\,i} + \mathrm{CALL}_{NN,\,i}. \tag{11}$$

Clearing (settlement).

$$\mathrm{CALL}_{NP,\,i}(X_m) = (X_m - S_{i-1})^+ - (X_m - S_i)^+\,, \tag{12}$$

$$= \min(\max(0, X_m - S_{i-1}),\; S_i - S_{i-1})\,, \tag{13}$$

$$\mathrm{CALL}_{NN,\,i}(X_m) = (X_m - S_i)^+ = \max(0, X_m - S_i). \tag{14}$$

Sign convention check: $S_i - S_{i-1} > 0$, so $\mathrm{CALL}_{NP,\,i}(X_m) \geq 0$ and is capped at $S_i - S_{i-1}$.

*[**TODO:** Split + clearing diagram for the bull call spread (zoom-in: $\mathrm{CALL}_N \rightarrow \mathrm{CALL}_{NP} + \mathrm{CALL}_{NN}$ at strike S_i).]*

*[**TODO:** Settlement chart: $\mathrm{CALL}_{NP,\,i}(X_m)$ vs. underlying USD price X_m (flat, ramp $S_{i-1} \rightarrow S_i$, flat cap $S_i - S_{i-1}$).]*

*[**TODO:** Settlement table for this tranche across X_m (the CALL_{NP} column of the combined table in Section 3.4).]*

3.4 Call-Side Conservation: Combined Chart and Table

After recursively splitting to S_n, the full call-side ladder identity is

$$X_m = \mathrm{CCR}_{S_0}(X_m) + \sum_{i=1}^{n} \mathrm{BCSR}_i(X_m) + \mathrm{CALL}_{S_n}(X_m), \tag{15}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \left[(X_m - S_{i-1})^+ - (X_m - S_i)^+\right] + (X_m - S_n)^+, \tag{16}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \min(\max(0, X_m - S_{i-1}), S_i - S_{i-1}) + \max(0, X_m - S_n). \tag{17}$$

The right-leg sum is telescoping:

$$\sum_{i=1}^{n} \left[(X_m - S_{i-1})^+ - (X_m - S_i)^+\right] + (X_m - S_n)^+ = (X_m - S_0)^+, \tag{18}$$

so $\min(X_m, S_0) + (X_m - S_0)^+ = X_m$: the tranches always sum back to the underlying.

Numerical check. Parameters $S_0 = 2000$, $S_1 = 2500$, with

$$\mathrm{CALL}_P = \min(X_m, 2000), \tag{19}$$

$$\mathrm{CALL}_{NP} = \min(\max(0, X_m - 2000), 500), \tag{20}$$

$$\mathrm{CALL}_{NN} = \max(0, X_m - 2500). \tag{21}$$

X_m	CALL_P	CALL_{NP}	CALL_{NN}	Total
3600	2000	500	1100	3600
2600	2000	500	100	2600
2500	2000	500	0	2500
2000	2000	0	0	2000
1700	1700	0	0	1700
1500	1500	0	0	1500
1000	1000	0	0	1000

*[**TODO:** Combined call-side chart: stack CALL_P, CALL_{NP}, CALL_{NN} as USD settlement value vs. underlying price X_m, showing they sum to the 45° line X_m.]*

4 Put Side

Begin with cash collateral $C = S_0$. At the base strike S_0 the cash splits into an obligation residual and a right:

$$C = (C - \mathrm{PUT}_{S_0}) + \mathrm{PUT}_{S_0}, \tag{22}$$

$$C = \mathrm{PUT}_{P,S_0} + \mathrm{PUT}_{N,S_0}. \tag{23}$$

The residual PUT_{P,S_0} is the cash-secured-put position (Section 4.2); the right PUT_{N,S_0} is the long put (Section 4.1), split further along the ladder into bear put spreads (Section 4.3). Unlike the call side, every put-side tranche conserves to the fixed cash value S_0, not to X_m.

4.1 Long Put (PUT_N)

How it works. The long put is the *right* half of the cash base split: a tokenized put position struck at S_0. Its holder owns the downside below S_0 and nothing above, giving short / bearish exposure to the underlying, while the matching residual (PUT_P) keeps the cash that collateralizes it. It is the only put-side leg that splits further.

Splitting. From the base split $C = \mathrm{PUT}_{P,S_0} + \mathrm{PUT}_{N,S_0}$, the long put is the right component

$$\mathrm{PUT}_{N,S_0} = \mathrm{PUT}_{S_0}. \tag{24}$$

Clearing (settlement).

$$\mathrm{PUT}_{N,S_0}(X_m) = (S_0 - X_m)^+ = \max(0, S_0 - X_m). \tag{25}$$

In cash-collateral units, with $C = S_0$:

$$\mathrm{PUT}^C_{N,S_0}(X_m) = \max\left(0, 1 - \frac{X_m}{S_0}\right) C. \tag{26}$$

*[**TODO:** Split + clearing diagram for the long put (zoom-in of the wheel: the* PUT_N *right leg).]*

*[**TODO:** Settlement chart:* $\mathrm{PUT}_{N,S_0}(X_m)$ *vs. underlying USD price* X_m *(*45° *down to* S_0*, then flat at* 0*).]*

*[**TODO:** Settlement table for this tranche across* X_m *(the* PUT_N *column of the combined table in Section 4.4).]*

4.2 Cash-Secured Put (PUT_P / CSPR)

How it works. The cash-secured put is the *residual* half of the cash base split: the cash minus the put that was sold off. Its holder keeps the full cash S_0 when the put expires worthless ($X_m > S_0$) and is converted into the underlying (now worth X_m) when the put finishes in the money ($X_m \le S_0$) — the cash-secured-put writer's position, used to acquire the underlying at a discount.

Splitting. From the base split,

$$\mathrm{PUT}_{P,S_0} = C - \mathrm{PUT}_{S_0}. \tag{27}$$

Clearing (settlement).

$$\mathrm{PUT}_{P,S_0}(X_m) = \mathrm{CSPR}_{S_0}(X_m) = S_0 - (S_0 - X_m)^+ = \min(X_m, S_0). \tag{28}$$

Settlement semantics: if $X_m \le S_0$ the put is assigned and the residual converts the cash into the underlying T; if $X_m > S_0$ it keeps the cash S_0. In cash-collateral units, with $C = S_0$:

$$\mathrm{PUT}^C_{P,S_0}(X_m) = \min\left(1, \frac{X_m}{S_0}\right) C. \tag{29}$$

Conservation with the long put:

$$\mathrm{PUT}_{P,S_0}(X_m) + \mathrm{PUT}_{N,S_0}(X_m) = \min(X_m, S_0) + (S_0 - X_m)^+ = S_0 = C. \tag{30}$$

*[**TODO:** Split + clearing diagram for the cash-secured put (the* CSPR *node: Cash*$\to T$ *when* $X_m \le S$*, keep Cash when* $X_m > S$*).]*

*[**TODO:** Settlement chart:* $\mathrm{PUT}_{P,\,S_0}(X_m)$ *vs. underlying USD price* X_m *(rising* 45° *to* S_0*, then flat cap at* S_0*).]*

*[**TODO:** Settlement table for this tranche across* X_m *(the* PUT_P *column of the combined table in Section 4.4).]*

4.3 Bear Put Spread (PUT_{NP} / BPSR)

How it works. Recursively split the put right at the next rung. For a put ladder with decreasing strikes

$$S_0 > S_1 > \cdots > S_n,$$

each higher-strike put splits into a capped bear put spread residual plus the lower-strike put that is carried further down the ladder.

Splitting. For $i = 1, \ldots, n$:

$$\mathrm{PUT}_{N,\,S_{i-1}} = (\mathrm{PUT}_{S_{i-1}} - \mathrm{PUT}_{S_i}) + \mathrm{PUT}_{S_i}, \tag{31}$$

$$\mathrm{PUT}_{N,\,S_{i-1}} = \mathrm{PUT}_{NP,\,i} + \mathrm{PUT}_{NN,\,i}. \tag{32}$$

Clearing (settlement).

$$\mathrm{PUT}_{NP,\,i}(X_m) = (S_{i-1} - X_m)^+ - (S_i - X_m)^+\,, \tag{33}$$

$$= \min(\max(0, S_{i-1} - X_m),\, S_{i-1} - S_i)\,, \tag{34}$$

$$\mathrm{PUT}_{NN,\,i}(X_m) = (S_i - X_m)^+ = \max(0, S_i - X_m). \tag{35}$$

Sign convention check: $S_{i-1} - S_i > 0$ (put strikes decrease), so $\mathrm{PUT}_{NP,\,i}(X_m) \ge 0$ and is capped at $S_{i-1} - S_i$.

*[**TODO:** Split + clearing diagram for the bear put spread (zoom-in:* $\mathrm{PUT}_N \to \mathrm{PUT}_{NP} + \mathrm{PUT}_{NN}$ *at strike* S_i*).]*

*[**TODO:** Settlement chart:* $\mathrm{PUT}_{NP,\,i}(X_m)$ *vs. underlying USD price* X_m *(flat cap* $S_{i-1} - S_i$*, ramp* $S_i \to S_{i-1}$*, flat* 0*).]*

*[**TODO:** Settlement table for this tranche across* X_m *(the* PUT_{NP} *column of the combined table in Section 4.4).]*

4.4 Put-Side Conservation: Combined Chart and Table

After recursively splitting to S_n, the full put-side ladder identity is

$$S_0 = \mathrm{CSPR}_{S_0}(X_m) + \sum_{i=1}^{n} \mathrm{BPSR}_i(X_m) + \mathrm{PUT}_{S_n}(X_m), \tag{36}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \left[(S_{i-1} - X_m)^+ - (S_i - X_m)^+\right] + (S_n - X_m)^+ , \tag{37}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i) + \max(0, S_n - X_m). \tag{38}$$

The right-leg sum is telescoping:

$$\sum_{i=1}^{n} \left[(S_{i-1} - X_m)^+ - (S_i - X_m)^+\right] + (S_n - X_m)^+ = (S_0 - X_m)^+ , \tag{39}$$

so $\min(X_m, S_0) + (S_0 - X_m)^+ = S_0$: the tranches always sum back to the cash collateral.

Numerical check. Parameters $C = S_0 = 2000$, $S_1 = 1500$, with

$$\mathrm{PUT}_P = \min(X_m, 2000), \tag{40}$$

$$\mathrm{PUT}_{NP} = \min(\max(0, 2000 - X_m), 500)\,, \tag{41}$$

$$\mathrm{PUT}_{NN} = \max(0, 1500 - X_m). \tag{42}$$

X_m	PUT_P	PUT_{NP}	PUT_{NN}	Total
3600	2000	0	0	2000
2600	2000	0	0	2000
2500	2000	0	0	2000
2000	2000	0	0	2000
1700	1700	300	0	2000
1500	1500	500	0	2000
1000	1000	500	500	2000
900	900	500	600	2000

*[**TODO:** Combined put-side chart: stack PUT_P, PUT_{NP}, PUT_{NN} as USD settlement value vs. underlying price X_m, showing they sum to the flat line S_0.]*

5 Wheeling (Reinvest Strategy)

Figure 1 renders the construction of Sections 3–4 as a state machine over the two collateral states—the underlying T and cash C. Each red entry arrow splits its state into an obligation residual ($\mathrm{CALL}_P/\mathrm{PUT}_P$, i.e. the CCR/CSPR residual) and a right ($\mathrm{CALL}_N/\mathrm{PUT}_N$); recall X_m is the terminal underlying price and S the relevant strike. At maturity the covered-call residual CCR converts the underlying into cash when the call finishes in the money ($X_m \geq S$) and retains T otherwise ($X_m < S$); its mirror, the cash-secured-put residual CSPR, converts cash into T when the put finishes in the money ($X_m \leq S$) and keeps cash otherwise ($X_m > S$). The right legs decompose recursively along the strike ladder into the bull/bear vertical-spread residuals BCSR/BPSR of Sections 3.3 and 4.3. This is the cycle that connects the two collateral states across maturities, and it is where time enters: each clearing re-mints collateral into the opposite state. The closed forms for every quantity labeling the graph are collected after the figure.

Equations referenced in the diagram

Base splits (one maturity step):

$$T = \underbrace{(T - \mathrm{CALL})}_{\mathrm{CALL}_P} + \underbrace{\mathrm{CALL}}_{\mathrm{CALL}_N}, \qquad \mathrm{Cash} = \underbrace{(\mathrm{Cash} - \mathrm{PUT})}_{\mathrm{PUT}_P} + \underbrace{\mathrm{PUT}}_{\mathrm{PUT}_N}.$$

Entry-split weights (token / cash units; here S denotes the base strike S_0 and X_m the terminal price):

$$\mathrm{CALL}_P = \min\Big(1, \tfrac{S}{X_m}\Big)\, T, \qquad \mathrm{CALL}_N = \max\Big(0, 1 - \tfrac{S}{X_m}\Big)\, T,$$
$$\mathrm{PUT}_P = \min\big(1, \tfrac{X_m}{S}\big)\, C, \qquad \mathrm{PUT}_N = \max\big(0, 1 - \tfrac{X_m}{S}\big)\, C.$$

Per-bucket spread and tail payoffs:

$$\mathrm{CALL}_{NP,\,i} = \min\big(\max(0, X_m - S_{i-1}),\, S_i - S_{i-1}\big), \qquad \mathrm{CALL}_{NN,\,i} = \max(0, X_m - S_i),$$
$$\mathrm{PUT}_{NP,\,i} = \min\big(\max(0, S_{i-1} - X_m),\, S_{i-1} - S_i\big), \qquad \mathrm{PUT}_{NN,\,i} = \max(0, S_i - X_m).$$

6 Summary

Compact identities

Call side, increasing strikes $S_0 < S_1 < \cdots < S_n$ (conserves to X_m):

$$X_m = \underbrace{\min(X_m, S_0)}_{\text{covered-call residual}} + \sum_{i=1}^{n} \underbrace{\min(\max(0, X_m - S_{i-1}), S_i - S_{i-1})}_{\text{bull call spread residual}} + \underbrace{\max(0, X_m - S_n)}_{\text{tail call}}. \tag{43}$$

Put side, decreasing strikes $S_0 > S_1 > \cdots > S_n$ (conserves to S_0):

$$S_0 = \underbrace{\min(X_m, S_0)}_{\text{cash-secured-put residual}} + \sum_{i=1}^{n} \underbrace{\min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i)}_{\text{bear put spread residual}} + \underbrace{\max(0, S_n - X_m)}_{\text{tail put}}. \tag{44}$$

Master tranche reference

Label	Formula / meaning
CALL_P / CCR	Covered call residual: $\min(X_m, S_0)$.
CALL_N	Call right (long call): $\max(0, X_m - S_0)$.
CALL_{NP} / BCSR	Bull call spread residual: $\min(\max(0, X_m - S_{i-1}), S_i - S_{i-1})$.
CALL_{NN}	Higher-strike tail call: $\max(0, X_m - S_i)$.

PUT_P / CSPR	Cash-secured put residual: $\min(X_m, S_0)$.
PUT_N	Put right (long put): $\max(0, S_0 - X_m)$.
PUT_{NP} / BPSR	Bear put spread residual: $\min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i)$.
PUT_{NN}	Lower-strike tail put: $\max(0, S_i - X_m)$.

*[**TODO:** Master worked example combining both sides + the wheel cycle across two maturities (extends the per-side numerical checks of Sections 3.4 and 4.4).]*

StrikeX Wheeling strategy

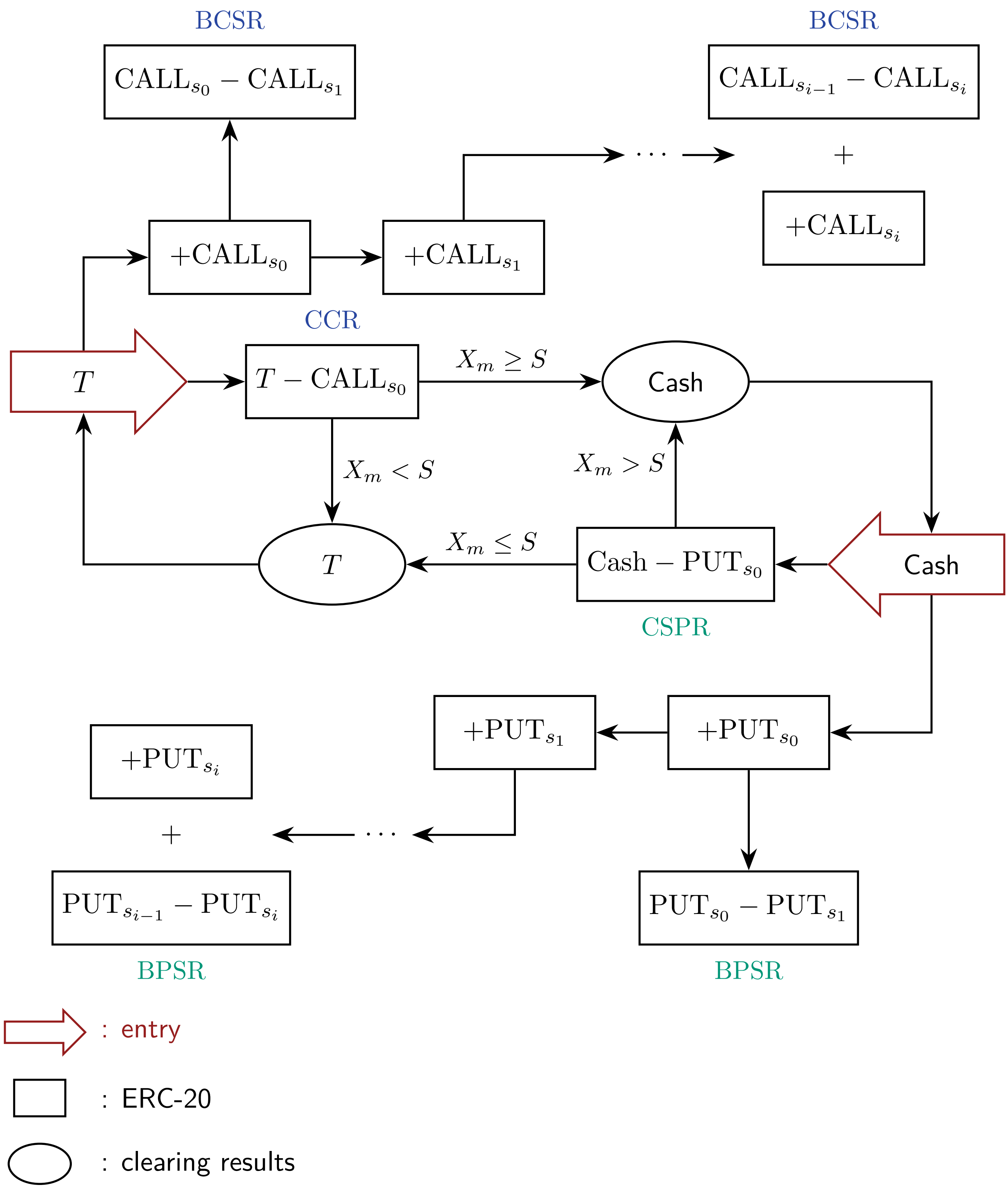

Figure 1: Tokenized option ladder as a recursion / state machine over the underlying state T and the cash state. Red arrows are entry points; each state splits into an obligation residual (P) and a right (N). The covered-call residual (CCR) and its mirror the cash-secured-put residual (CSPR) move probability mass between the T and Cash states at maturity according to X_m vs. the strike S, while the right legs recurse along the strike ladder into bull/bear spread residuals (BCSR, BPSR). This figure is the visual counterpart of the algebraic identities in Sections 3–4.